ONE SIZE DOES NOT FIT ALL! GENDER IN/EXCLUSION IN A RURAL COMMUNITY-BASED ICT INITIATIVE

Wendy Faulkner and Tine Kleif
University of Edinburgh

ABSTRACT

Community-based information and communication technologies (ICT) networks are seen as an important means of reducing social exclusion, and at the same time fostering community development. Increasing ICT capability locally is arguably crucial to furthering both of these broad aims. This paper presents evidence about the capability-building activities and impact of one community-based ICT initiative, which we have called the Ardmore Network. This is a network of part-time staffed Community Resource Centres in a remote rural region of Scotland. It provides three services: access to informational and technological resources, one-to-one support in learning to use these resources, and basic and vocational ICT training. The initiative is indeed increasing ICT capability amongst sections of the community. Our analysis identifies two keys to this success, which we believe have salience for other community-based efforts to increase ICT capability. First, it involves a *heterogeneous* package of measures in the sense that it is not 'just' about providing public-access computers. And second, it works to the degree that it facilitates *informal* ICT learning through networks of 'local experts' within the community. However, the Ardmore Network attracts more women than men, and is failing to reach many non-ICT-using men in the communities it serves – especially middle-aged men in manual work – in spite of being promoted 'for everybody'. We identify aspects of the 'one size' strategy which do not 'fit' this section of the community, and conclude that digital inclusion efforts must be based on gender-aware effective targeting if they really are to include everybody in the community

Keywords: ICT capability, rural communities, gender, digital inclusion, informal networks

Introduction

In November 1999, the Scottish Executive published its vision for tackling poverty and injustice in Scotland and set milestones for the creation of a more inclusive society. One of the disadvantaged groups identified was people living in remote communities, and one of the milestones was to reduce inequalities between communities by increasing access to information and communication technologies (ICT). Accordingly, a range of community-based ICT initiatives has since been implemented in Scotland.[1] The paper examines one such initiative, which we are calling the Ardmore Network. It comprises a network of part-time staffed Community Resource Centres across an extensive, remote rural region, and has had notable successes in facilitating the uptake of ICTs amongst both individuals and community groups. Like others of its type – and indeed like many digital inclusion strategies – this initiative was intended for all members of the community. Yet, as we will show, 'for everybody' community-based ICT initiatives don't always reach 'everybody'. Specifically, as in this case, they may be 'gender blind'.

Community-based ICT initiatives – variously referred to in the language of 'wiring communities up' (Schuler, 1996), building 'community information networks' (Pantry, 1999) or 'community informatics' (Gurstein, 2000) – are supported on several grounds.[2] First, a major driver behind all digital inclusion strategies is the belief that nations must build the skill base and the infrastructure necessary to take advantage of, if not be at the forefront of, continuing techno-economic development of ICTs (e.g. the 2000 reports by the UK Policy Action Team 15, and the Digital Scotland Task Force). Second, a connection is often drawn between digital exclusion and social exclusion, arguing that if the more socially excluded groups are not effectively included in the information society they will be further disadvantaged in the future (Day and Harris, 1997; Clarke, 2002). Making ICTs cheaply available locally and providing appropriate training is being promoted as a means of increasing public use of the technologies in disadvantaged communities. Thus, locally based ICT projects have been increasingly prioritised within programmes like the Social Inclusion Partnerships in Scotland as a means of simultaneously closing the digital divide and tackling social exclusion (Malina, 2001).[3] Third, such projects are also seen as supporting community development and citizenship. In this connection, 'ICT is frequently viewed as a democratising force' (Malina, 2001). ICT initiatives thus have a strong emphasis on being user-led rather than technology or expert-led (Schuler, 1996), on making ICT resources available to grass roots activities and on nurturing the widest possible democratic participation (Ascherson, 2003). ICT is also frequently viewed as facilitating decentralised modes of communication, thus having the potential to strengthen communication and networks – both within and between communities (Schuler, 1996). Fourth, and linked to this, community-based ICT initiatives are often supported as a means of enhancing the delivery of public services, on the grounds that ICT provides the potential to render these services more convenient, more citizen-centred and more efficient.

Community-based ICT initiatives or networks may vary on a number of counts (Day and Harris, 1997; Pantry, 1999; Holligan, 2003): the origins and funding (local authorities, private sector, users: Miller, 1999), the geographic character of the community (neighbourhood, city, region), the topic or interest defining the network (educational, interest-based or general), and the location of the ICT resources (schools, business, public spaces). But in all cases, community networks are charac-

terised by an emphasis on 'local relevance', on access and inclusion, and on community involvement and development. And in most cases, there is an emphasis on providing training and support so that everyone has the potential to benefit from the new technologies. Indeed, we would argue that increasing ICT capability locally is critical to the potential of community-based ICT initiatives to further either social inclusion or community development aims. We use the term capability here to indicate not only the *skills* required to use a range of ICT applications, but also the capacity to make *judgements* about how and when to use (and potentially adapt) ICTs for particular purposes, plus the *confidence* to do so.[4] Accordingly, this article concentrates on how the Ardmore Network has contributed to enhancing ICT capability locally; it does not assess its contribution to the underlying aims.

Unlike other studies of community-based ICT initiatives,[5] we chose to investigate the Ardmore Network through an interest in *gender* inclusion in the information society – and gender and ICT more broadly. The research was conducted as part of a collaborative European project entitled 'Strategies of inclusion: gender in the information society' (SIGIS), which sought to investigate a range of public and private sector strategies to include more women in the design of use of ICTs.[6] The focus on women (*qua* gender) in this project was justified by evidence of a continuing (though diminishing) digital divide between men and women in terms of access to and use of ICTs, and of a continuing predominance of men amongst ICT specialists (Sørensen and Stewart, 2002). Previous work suggested that part of the explanation for this gender divide was the existence of residual cultural associations between technology and masculinities (Lohan and Faulkner, 2004), and in particular associations of computer enthusiasm with men and computer reticence with women (e.g. Håpnes and Rasmussen, 1991). The Ardmore Network was interesting to us because, counter to these conventional expectations about gender and ICT, it attracted and involved more women than men, including many previously non-ICT-using, computer-reticent women. Ironically, given the framing of SIGIS, this 'for everybody' strategy was 'gender blind' in so much as it failed to reach many non-ICT-using and computer-reticent *men* in the locality.

Following from the above, our concern in this paper is twofold: first, to illustrate how community-based ICT initiatives can be effective in increasing ICT capability across remote rural communities; and second, to argue that such digital inclusion strategies must be based on gender-aware effective targeting if they really are to reach 'everybody' in the community. In the following section, we describe the Ardmore Network strategy and its success in increasing ICT capability locally. We then identify two elements in this success which we believe have salience for other community-based efforts to increase ICT capability. First, the initiative involves a 'package' of measures which is *heterogeneous* in the sense it involves much more than just providing public-access computers. Second, it works to the degree that it facilitates *informal* ICT learning through networks of 'local experts' within the community. The next section analyses the gender and other factors influencing who does and does not use the Ardmore Community Resource Centres. This demonstrates not only that more men than women are digitally excluded in the region, but also that certain aspects of the inclusion mechanisms miss, and so exclude, computer-reticent men. Our conclusion points to ways of overcoming this. First, we outline the methodology adopted for the case study.

As indicated earlier, our investigation of the Ardmore Network was designed to provide a case study of successful digital inclusion of women; in no way does it

constitute an evaluation of the initiative. Data collection occurred in the summer of 2002, combining semi-structured interviews with analysis of relevant documents.[7] We analysed the network's website, promotional material and annual reports, as well as relevant information from Scottish Executive websites, in order to establish the stated aims, achieved objectives and marketing messages of the initiative. We conducted interviews with six staff from the Ardmore Network – five of the seven Community Support Workers plus the Project Development Worker. These interviews provided a reasonably rounded picture of the practices and uses of the network and its Resource Centres. Nine further interviews were conducted, with four women who use the centres and five men who do not – all identified and contacted on the basis of 'snowballing'. These interviews enabled us to triangulate and extend the staffs' accounts of the Ardmore Network to some extent. Most crucially, they provided valuable insights into why the initiative proved to be more successful in reaching women than men – albeit our data about use and non-use are necessarily suggestive rather than systematic because of the scale of the study. All of the interviews lasted around one hour, and were recorded either on tape or with detailed notes. In presenting the material we have anonymised the places and people involved; the particulars are not important for the points we make in this article. We believe that the Ardmore region is typical of other remote rural regions in the Scotland (and possibly elsewhere), and that our main conclusions – concerning how to improve ICT capability locally, and the need for gender-aware targeting of such measures – are relevant and applicable to community-based ICT initiatives in general.

The Ardmore Network strategy

Ardmore is an extensive rural region, remote from the main urban centres of Scotland. Typical of such regions, it struggles to sustain employment from tourism, forestry and fish, and has a relatively large retired population. Some 21,000 people live in the region: roughly half in its only town, the remainder scattered across a vast hinterland. Ardmore thus contains a variety of localities; some have pockets of urban-type deprivation and others are extremely isolated. Remoteness has knock on effects for all aspects of economic and social life – and, therefore, in/exclusion. Obtaining supplies for the home or business can be very difficult and expensive when even travelling to the region's town is a lengthy and costly journey. The sense of isolation can be debilitating, especially in winter.

The Ardmore Network was set up in 1997 and won three years funding from the Scottish Executive Pathfinder Initiative in 1998 to establish a network of Community Resource Centres across the region. The Pathfinder remit was to use ICT to develop electronic means of delivering services in rural localities, so overcoming some of the problems associated with remoteness. The aim of the Network was 'to empower communities in Ardmore through access to communications and information', which encouraged links with other social inclusion and community development efforts from an early stage. The nine Community Resource Centres are located in a range of communities – a public housing scheme in the main town, semi-remote medium- and small-sized villages, and remote villages – all but the first of which also cover large surrounding areas. Typically, the Community Resource Centre is a room in an easily accessible building such as a local community centre. Each is equipped with various technologies (a computer, software, printer, scanner, digital camera) and information (e.g., on fund raising). Each is staffed 12 hours a week by a Community Support Worker.

The Community Support Workers – there are seven in total – are responsible for promoting the use of the Ardmore Network, maintaining the equipment, providing one-to-one support and identifying training needs. They also have a wider remit to support community development (e.g. through community planning) and facilitate user participation, and can work some additional hours to meet these objectives. Indeed, they were chosen for their local knowledge and their potential to contribute to community development, rather than for their ICT skills. None of them has specialist ICT qualifications or work experience. Those we spoke to described themselves as 'competent rather than expert' ICT users; most were completing the European Computer Driving License (ECDL) at the time. A central Project Development Worker (qualified in Community Development) is responsible for supporting and coordinating the efforts of the Support Workers, for setting up and promoting the Network website, and for publicity and fund-raising for the network. At the time of our study all of the Ardmore Network staff are women, ranging in age from 30s (two only) to 50s. All but one of the Community Support Workers to whom we spoke are also mothers.

The Ardmore Network is explicitly promoted for all members of the community, irrespective of age, gender, employment or level of computer expertise. Marketing is a major responsibility of the Community Support Workers, who place adverts in local newspapers, put up posters, hand out leaflets and raise awareness of the centres through their own local networks (most importantly, bumping into people at the village store) and outreach efforts (e.g., to local mother-and-toddler or youth groups and through other community activists). Where the Support Workers are local (most cases) word of mouth is more effective than advertising in getting people through the door. The message throughout is that ICT is interesting, useful, fun and not scary; the centres have something to offer you whether you know about computers already or not, whether you want to use the technology to have fun or do serious tasks, and so on. There is a strong 'something for everybody' message in all the publicity, formal and informal.

The Community Resource Centres offer three main services, to individuals and community groups alike:

- use of the technological and informational resources
- one-to-one support in using these resources
- basic and vocational ICT training.

Registered users are set up with an email account and a personal folder on the computer; they are able to use the resources at any time. Casual users may use the resources only when the Community Support Worker is present.[8] Some of those who make use of the centre's resources already have computers of their own and/or are ICT literate. But not all users know what to do: the one-to-one support is intended for them. They might need support to do a specific task, such as producing a CV, downloading photographs or searching the net, or they might need support to get started with ICT at all. Many of these users would not have had the confidence to go on a course without this initial encouragement (see next section). The basic ICT training courses are all short: usually one-day workshops or 2 or 3 evening or daytime sessions, depending on what people want. They are organised by the Community Support Workers and delivered, to groups, either by them or by staff from the local college. Some are introductory for complete beginners (along the lines of 'computers for the

terrified'); others offer training in specific applications (databases, web-page design and so on). The vocational ICT training available is the more committing ECDL, which has to be completed within three years. This is offered at a subsidised fee and delivered as distance learning, supported by a tutor from the local college.

The Ardmore Network has had some tangible benefits. With respect to community development, the Community Resource Centres provide resources (e.g., to produce publicity materials) and a place to meet; the Community Support Workers also facilitate community action, principally through fund-raising for community projects, help with setting up websites and networking between community groups across the region.[9]

The main benefits of the Community Resource Centres to *individual users* reported to us are indicated below.

Reducing the need for travel, through on-line services

As intended, many of the uses made of the Ardmore Network's ICT resources, in all localities, do indeed help overcome some of the disadvantages of remoteness, and reduce the need for long and expensive journeys. The use of the Internet has served to 'shrink distances' and 'make life easier' for people living in remote areas – by providing on-line access to information (e.g., about health, travel, jobs), email, shopping and other services.

Being able to use technologies which would be too expensive to buy

This is a benefit for community groups and individuals alike. It often means that groups can do things which would not otherwise be possible – such as producing digital pictures of a recent mother and toddlers' outing, or developing a website to promote local tourism. Many individuals who cannot afford their own on-line facilities rely on the Community Resource Centres for access to the Internet.

Keeping closer in touch with distant family and friends

For elderly residents with far-flung family, for younger residents who have distant friends, and for visitors to the region, the email facility of the Community Resource Centres is highly valued.

Having fun and enhancing leisure pursuits

In all the Community Resource Centres the digital camera is the most frequently and widely used piece of technology, which serves to remind how having fun with technology can act as a great facilitator for the uptake of ICTs. In a similar vein, hobbies often provide a motivation to gain computer skills in order to access resources from the Internet or to get in touch with people who have the same interests (see below).

Keeping up with technology

Many people are motivated to gain basic ICT skills to help them keep up with their kids or grandkids, or to benefit from all the other 'making life easier' uses of computer-based technology. There is a tremendous demand for this training: In the

year 2001/2, the Ardmore Network had aimed to provide 250 basic training places through the Community Resource Centres: in the event nearly 700 were taken up.

Gaining labour market skills

Most of the communities in the region have limited employment opportunities, especially those women who get stuck in seasonal low-paid jobs in tourism and 'don't think they can do anything else'. This explains why most of the nearly 50 residents who took the vocational ECDL training are working-age women. Whilst there are few office jobs outwith the main town, several of these trainees have put the ICT skills gained to good use in various small businesses locally.

Building self-confidence

Many users, especially those with a low sense of their own worth, experience a great boost to their confidence and self-image through acquiring ICT skills. One support worker described a group of lone parents she has trained as 'brimming with self-confidence' after their achievements on the computer. Another spoke of some retired residents who she has supported one-to-one: 'People think "Well I can't do that, I am too old or too scared", and then they find they can. So it is a great confidence-builder.' In this as in other ICT training initiatives,[10] confidence building can prove a pivotal link between digital inclusion and social inclusion.

The Ardmore Network is clearly effective in reducing some of the disadvantages of remoteness in a rural community, and in empowering (in a variety of ways) residents. We note that crucial to realising all of these benefits has been the Ardmore Network's success in increasing ICT capability amongst many groups in the community, especially through the basic ICT training and one-to-one support. This has been identified as a strong feature of the network's activities in both of its two evaluations. We now turn to discuss the two aspects of the Ardmore Network which we identified as contributing to its success in this area.

A heterogeneous package of measures

One reason why the Ardmore Network initiative is effective in increasing ICT capability locally is because it involves a *heterogeneous* package of elements. We use the term heterogeneous in the sense coined by sociologists of technology (e.g., Bijker *et al*, 1987) – that technological and social elements are necessarily interdependent; both are densely interwoven throughout the social fabric. Thus, the benefits listed above all serve to remind that it is not the technology *per se* which is inclusive but the things people can use it for. Similarly, the Ardmore Network initiative is successful because it is *not just* about putting computers in public places – this alone does little to enhance ICT capability locally – but because the technology is backed up by a number of other 'social' measures.[11] Most significant of these measures, is the involvement of the Community Support Workers who promote its relevance to people's lives and facilitate them gaining the necessary ICT capability to take full advantage of the new technologies. The Community Support Workers are effective because they bring commitment, enthusiasm and local knowledge to the job; their ICT capability is a necessary but not sufficient condition. Because they are connected within the community, they can promote its

use effectively and provide a trusted bridge, a familiar face for those who might feel wary of computers.

Similarly, the Ardmore Network's efforts at capability-building are successful because they are *not just* about formal training; this is also backed up by informal one-to-one support. The one-to-one support is especially important in building up ICT capability amongst the computer-reticent or generally unconfident, who may have had bad experiences at school or around ICTs and feel uncomfortable about embarking on formal ICT training. For example, one woman who came in had not used her computer for months because her husband once shouted at her for not shutting it down properly. He told her she had broken it and she believed him, so lost all confidence about going anywhere near her computer. Two Community Support Workers describe how they tend to work in their initial contact with reticent users:

> *They don't want a course. I wouldn't even call it training – that is too big a word sometimes. It is 'what would you like to do?', and going from that to expanding themselves and maybe even thinking about work, and even looking up jobs, and everything.*

> *Sometimes we started completely informally and just got people to look at a shopping website or some health issues for their children or diabetes. Just tried to get chatting about anything that interested them and say 'why don't you come and have a look at the website? You don't have to do anything.' Just to try and start an interest off.*

One-to-one support works partly because it creates a safe, comfortable environment, and partly because the support can be tailored to the individual user, delivered to meet their particular interests and needs, and at a pace that suits them. One retired woman user met with the Community Support Worker over five weeks before she felt she could use Excel without help to do her household accounts. Another, Dorothy, was 'terrified' of computers before she was persuaded to come on a basic training course which opened a whole new world for her. She and her best friend from the local art group, now attend the centre every Tuesday afternoon to 'visit' art galleries around the world on the Internet, and to produce artwork using PhotoDraw. By such routes, ICTs become part of people's everyday lives.

All of this – effective outreach, creating a safe environment, and tailoring individual support and training to people's interests and needs – helps tackle a lack of confidence or self-esteem. In addition, the Community Support Workers provide an accessible role model for unconfident users in terms of their approach to ICT. In the course of one-to-one sessions, they frequently stress their lack of specialist ICT qualifications or work experience to users but demonstrate, in their interactions with the computers, that this does not have to be a disadvantage. When they don't know how to do something they will 'have a go' *with the user present* until they have worked it out together – a very positive demonstration of what an 'ordinary' user can achieve if they are confident enough to experiment.

Getting the computer-reticent and the unconfident through the door, making them feel comfortable about coming to the Centre, and building their trust to the point where they can become confident ICT users, all takes time and effort. The fact that it takes such 'patient labour' to be effective with those who are reticent and unconfident – who arguably have most to gain from the Ardmore Network – indicates a need for

sustained revenue funding. However, the continuing rhetorical support for lifelong learning and ICT capability-building does not seem to translate into ongoing financial support for existing projects like the Ardmore Network, which exist in a very unpredictable funding environment. This has been a recurring complaint about community networks (e.g. Harris and Dudley, 2000).

Informal networks of local ICT experts

The second key to understanding why the Ardmore Network is effective in increasing ICT capability – and indeed to understanding why it is effective with particular social groups and not others (next section) – is the crucial role played by informal learning and networks of local experts in ICT capability-building.

Numerous studies have together established the following general points. First, informal learning typically takes place alongside, and supplements, formal learning about ICT. Formal ICT training for teachers, for example, is most likely to be effective when they spend time practising and developing new uses of ICT in their own time at home (OFSTED, 2002). Second, in a study of how people from all walks of life incorporate ICTs into their everyday lives, Stewart (2001) demonstrates how, in the course of such informal learning, people almost invariably draw on one or more *local ICT experts*. These are not generally experts in the formal sense, but someone people know from their circle of family or friends, to whom they turn for help and advice. Such local experts often act as positive role-models for building confidence and enthusiasm about using computers. Third, people's life settings – their age, work and social networks – all have a direct bearing on what form their local networks of ICT expertise takes (see next section). Fourth, such networks are not static but contribute dynamically to the build-up of ICT capability. As one local expert's capability increases over time, so does the potential capability of others in that person's social networks: there is a multiplier effect.

It follows that the presence and dynamism of networks of local ICT experts will facilitate informal learning and, thus, the building of ICT capability within communities. This is well-illustrated by one Community Support Worker who had played the role of informal local ICT expert in her neighbourhood before working for the Ardmore Network – encouraging others to become more competent ICT users and helping them with any technical problems. Over time, a network of such local experts has developed locally as a result of her efforts. Where no-one in her street used to own a computer, there are now three people who do. 'The cover is always off' these computers because their owners are always exchanging parts or testing things out. Old computers get passed on to neighbours, who in turn come to these local experts if they have a problem. In this way, the network has had a snowball effect on the general level of ICT confidence and competence in the village. People know who they can ask for help, but many of them also know a lot more about computers now themselves.

To explore informal ICT learning and experts further, we asked both users and non-users of the Ardmore Network to say who got them started on computers, and who they turn to when they need advice or help with ICT. Through their answers it was easy to identify local experts and networks. In all cases these emanate from *existing* social networks. Thus, for example, the local ICT expert for two young ICT users we interviewed is the computer enthusiast parent who got them going in the first place, but each of these young users now fills the role of local expert for their

circle of friends – hence the network analogy – and each has been instrumental in increasing the ICT capability of their friends. Inevitably, some groups have fewer local ICT experts in their social networks than others. Mharie and her husband, for instance, were first persuaded to buy a home computer by their computer-enthusiast son, and relied on his help with ICT, but had no one to turn to when he went away to university. Being middle-aged, they are less likely than those who learnt about ICT at school or to have local ICT experts amongst their peers, unless their circle of friends includes people in occupations where they need to use computers (see next section).

In sum, informal ICT learning through networks of local experts helping out and passing on their knowledge and enthusiasm to others who have less expertise or confidence, is clearly very important in ICT capability-building – and by no means unique to remote rural communities. We believe this conclusion is of huge policy significance for digital inclusion efforts. Arguably, community-based ICT initiatives like the Ardmore Network Community Resource Centres have an especially important potential contribution to make in relation to those groups (e.g. many of the over-30s) who have relatively sparse informal networks of ICT expertise.[12] Time and again, we heard of Community Support Workers acting as role models and local experts for older residents, and of how this facilitates informal learning more widely in the community. For example, Dorothy's best friend would never have known how much computers could bring to her interest in art and drawing if Dorothy had not been persuaded to overcome her fear of computers and come along to the centre in the first place. In the next two sections, we consider why this approach has not worked for the 'missing men'.

Digital in/exclusion: gender and other factors

As noted in our introduction, the Ardmore Network Community Resource Centres are used by more women than men – in spite of being intended and promoted for 'everybody' in the community. The range of actual users, and the patterns of usage, vary somewhat across the different centres – reflecting in part pre-existing differences between the localities, and in part the particular social networks and outreach efforts of individual Community Support Workers. For all these differences, however, the majority of those *individuals* using the resources and benefiting from the one-to-one support and training offered are women in all cases: ranging from 60 per cent to 80 per cent in the centres we studied. (Community group usage seems to be more gender-balanced.)

There are distinct patterns of individual usage for different groups of women and men. Thus, the basic ICT training is followed predominantly by retired aged women and (some) men, while the vocational ICT training is followed predominantly by working-age women. Very few men take advantage of the one-to-one support available through the centres: the main users of this service are retired women and unemployed mothers, including lone parents, many of whom are computer-reticent and/or generally lacking in confidence. Put another way, the Ardmore Network is particularly successful in attracting and building the ICT capability of four groups of women users:

- (mostly young) computer-skilled and enthusiastic women who use the centres for cheap and easy access to ICT resources which they can't afford to own themselves;

- young mothers, including lone parents, who want to 'keep up with their children';
- working-age women who want to improve their position on the labour market;
- retired-age women who want to stay in touch with distant family or pursue hobbies and other interests using ICTs.

Largely absent amongst the users of all the Ardmore Network Centres, unless involved through various community groups, are men of working age.

These findings were obviously interesting to us in relation to strategies for gender inclusion in the information society. By interviewing a selection of women and men from the region, we sought to understand better why relatively many women and relatively few men make use of the Community Resource Centres. We compiled a modest 'techno-biography' (cf. Henwood, Kennedy and Miller, 2001) for each informant, which means we tried to build up a picture of the history of their use/non-use of ICTs, and to make sense of this in the context of wider aspects of their lives and identities. We also asked our interviewees about ICT use and capability amongst their circle of friends and family. Together with evidence from the staff, this shed light on their use/non-use of the Ardmore Network Centres.

Digital in/exclusion in Ardmore is shaped by the same factors as elsewhere: age and occupation are key determinants of ICT literacy and use, with gender cutting across (Sørensen and Stewart, 2002). Those under the age of 30 have acquired basic ICT competence at school, whilst those over 30 have generally learnt about computers (if at all) because they needed to for work. For two women and one man under 30, an added factor was the presence of a computer-enthusiast parent who introduced them to computers early on and acted as a positive role model.[13] Gender shapes ICT literacy and use to the extent that employment and post-school education are gender-differentiated. Thus, most of the men in Ardmore have permanent manual jobs in forestry, fisheries or fishing, where the women work mostly in seasonal jobs in tourism, or in offices. Women are more likely than men to work in jobs where computers are used. And, more women than men from the region go on to university (where computer use is required).

All of this points to greater digital exclusion amongst men in the Ardmore region than amongst women, especially amongst men over 30 years of age. This is well illustrated by the contrast between two of the women we spoke to and their male partners. Susan is a university graduate in her 20s who 'couldn't live without a computer' because email is how she now keeps in touch with her university friends. By contrast, her boyfriend has basic computing skills but isn't interested in using them; he doesn't need computers in his work and has never left the locality so doesn't have distant friends. Mharie is middle-aged; she picked up ICT skills through previous jobs in small businesses and is now doing the ECDL training. By contrast, her husband has not pursued formal ICT training even though he has some ICT skills and is currently out of work. She says he is unconfident about computers and 'is not ready to admit that he doesn't know everything about it'.

Our study identified other such computer-reticent men as well as computer-enthusiastic women, counter to the conventional assumptions about gender and ICT outlined in our introduction. Of course, there is considerable diversity amongst the male population, as amongst the female. Some men in the region are already very ICT-literate and have all the equipment they need at home. A larger proportion do not need ICT skills for work. Some of these simply are not interested, like Susan's boyfriend. Others report that they do not have time to learn more about computers (e.g.,

because of a long spell out at sea). Those men who are young enough to have learned basic ICT skills at school generally have at least some computer enthusiast friends they can turn to should they need to. Arguably the most worrying group, in terms of inclusion in the information society, are those men over 30 years old who, like Mharie's husband, have not had the opportunity, at school or at work, to learn about ICT and who are far less likely to have 'local experts' (see below) amongst their friends. It seems likely that gender cultures and identities may be operating in addition to the obvious structural factors of the gender segregation of labour (cf. Harding, 1986). Thus, men whose identity is strongly wrapped up with working in 'men's jobs', and who socialised principally with other men, may be put off computers because they associate them with women's work and with education. It also seems likely that many such men will, like Mharie's husband, tend to be reluctant to expose their ignorance about ICT to others.

Gender in/exclusion in the Ardmore Network

That more men than women are digitally excluded in the region is only part of the explanation for the relative failure of the Ardmore Network to reach these computer-reticent men. Whilst not all non-users of ICT are 'a problem' (Wyatt *et al*, 2002), there can be no doubt that many men could benefit greatly from the initiative. The key question is: why aren't the Ardmore Community Resource Centres attracting and working for men non-users, especially the more computer-reticent men, when they are clearly working for various groups of digitally and/or socially excluded women? Our case study suggests that, there are four aspects of the Ardmore Network strategy which are in practice more effective in relation to women than to men:

The marketing of the Ardmore Network does not reach many men

As noted earlier, the marketing efforts of the Support Workers rely heavily on word of mouth and thus on their own social networks. To the degree that social networks in the region are gender-segregated and the Community Support Workers are all women, their efforts are bound to miss at least some men. Every Community Support Worker mentioned the importance of meeting people in the local shop, but not the pub. And many told us they try to reach the men through their wives. As Harris and Dudley (2000) report in a study of Learning Centres, word of mouth does not always reach socially excluded groups. Further, to the degree that the Ardmore Network's outreach efforts target organised community groups, they are not going to reach those men who do not participate in such groups; the emphasis on youth groups and young mothers is clearly missing older men!

The Resource Centres are perceived and/or experienced as 'women's spaces'

Our evidence suggests that the social spaces and cultures existing in the Ardmore Network Centres feel like 'women's spaces' (cf. Laurie *et al*, 1999). All the women users we spoke to find the Community Resource Centres comfortable places to visit. Some mothers report that they use the centres even though there is a computer at home because they can concentrate on the computer there without the distractions of home. Lone parents and retired-age women especially value the social contact which

comes with visits to the centres. The Community Resource Centre can also have this effect for younger people. Rosie (16) is a not very socially confident teenager. She visits the centre because she can't afford Internet access on her own computer and because she finds the atmosphere there friendly. She brings her friends along to show them the amazing things they can do and find out about on the Internet. By contrast, Jim (32), who is also unemployed and suffering from depression, is not ICT-literate. Even though he would like to learn about computers, and knows the staff at his local Community Resource Centre, he says he and his (mostly younger) friends would not visit the centre;

> *They can find out everything they want from their friends a lot of the time. They would go to their friends who know a lot about computers rather than go down there, you know? […] I think it's because they can speak to their friends openly, in a way that they understand, really. Obviously, if you can go to your friends about some information about the computer, you would go to your friends rather than go down there.*

It seems 'obvious' to Jim that his friends would rather expose their ignorance to each other than to the middle-aged women Support Workers at the Community Resource Centres!

The Community Support Workers are not appropriate role models for all groups

Following from the above, it is clear that the same Community Support Workers who provide such positive role models for ICT use amongst those women who lack self-confidence or are particularly wary of computers are not playing this role for computer-reticent men. It seems likely that such men may be more willing to expose their ignorance about ICT with a male Support Worker.[14] Certainly, there is evidence that same-sex role models work for both women (Faulkner and Kleif, 2003) and men (McKeough, 2003) in basic and vocational ICT training. Gender may not be the only issue here. In the case of the Ardmore Network, the support workers all very similar on several counts – they are mostly middle-aged mothers. Arguably, there is a case for having training and support provided by a more diverse range of staff – reflecting differences in age, circumstances and background as well as gender.[15]

Adult education is not something that male manual workers generally do

The gender split amongst individuals taking advantage of ICT training and support in the Ardmore Network is in line with a general and long-standing pattern in community-based adult education whereby classes are dominated by women, especially in the over-40 age group, and the most under-represented group of men are manual workers and unemployed (McGivney, 1998, chapter 1). McGivney (chapter 3) explains these missing men by a number of factors which resonate with the Ardmore Network evidence: a version of masculinity tied up with work and male peer-groups which carries anxiety about 'losing face'; a perception that adult and continuing education is for women, partly because it is mainly staffed and attended by women; and a reluctance to admit learning needs (see also Tett, 1994). Interesting in our context, it is often asserted that, since the range of subjects available in adult

education is generally more interesting to women than men, one way of attracting men is to offer courses in ICT (e.g. Midgeley, 1997)!

Conclusions: more effective gender targeting of 'for everybody' initiatives

In many crucial respects, the Ardmore Network represents a good model for community-based ICT initiatives. In particular, its success in building ICT capability locally can be attributed to the heterogeneous package of measures involved, especially the provision of part-time Community Support Workers alongside the technology, who can act as role-models and local ICT experts for those who are computer-reticent and/or do not have many opportunities for informal ICT learning in their own social networks. However, the 'one size fits all' strategy of the Ardmore Network has not succeeded in engaging all sections of the community equally. Like one-size clothing, the strategy does not 'fit' some people – most notably in this case, working-age men, including many non-ICT-using computer-reticent men.

We believe the Ardmore Network story has wider salience in a context where many community-based ICT initiatives may be hoping to reach 'everybody' but actually missing *significant and identifiable* sections of the community. As McGivney notes in relation to adult education,

> *If we have education and training programmes and institutions which are ostensibly open to all but which attract only certain segments of the population, then that 'openness' is itself open to question.* (1998, p 1)

We have demonstrated how various aspects of the Ardmore Network strategy are gender marked: the marketing, the social spaces of the Resource Centres, the predominance of women Community Support Workers, and a long-standing association of adult education with women. Clearly a more diverse range of *effectively tailored* digital inclusion efforts is needed if all sections of a community are to be reached. The Ardmore Network case highlights a crucial further conclusion of our study: in order to achieve this, it is vital to be aware not only of how *exclusion mechanisms may be gendered*, but also of how inclusion mechanisms may be gendered. The case may seem self-evident, but it bears stressing in a context where digital inclusion initiatives are frequently 'gender blind' (e.g. Faulkner and Stewart, 2003). It also bears stressing that such gender awareness must be based on an understanding of the diversity *amongst* men and *amongst* women, rather than assuming that all women are necessarily one thing and all men necessarily something else.[16]

Several propositions arise from our study of the Ardmore Network. In particular, it suggests that if community-based ICT initiatives are really to reach everybody, including the missing men, they need to attempt the following:

- To think about which social groups are likely to be least included in the information society, and to acknowledge the particular factors shaping ICT non-use by different groups, including any gender related factors.
- To locate and nurture existing local ICT experts in order to facilitate informal ICT learning, especially amongst social groups and networks where this is rare – such as amongst middle-aged men in manual jobs.
- To reach out to the missing men *in their existing social networks and spaces* (as

happens in the Ardmore Network with young mothers); many outreach initiatives have taken adult education to the pub, for example (McGivney, 1998, chapter 5 and 2000).
- To create learning settings where the computer-reticent might be more willing to expose their ignorance; it is reported that one men-only course on ICT, for example, overcame men's anxiety about exposing their ignorance in front of women who have better keyboard skills than them (McGivney, 1998, chapter 5).
- To consider ways of making community network spaces more comfortable to all, and to use a diverse mix of trainers and support workers who reflect the diversity of community they serve.

Acknowledgements

We are grateful for the financial support of the Information Science and Technology section of the European Union for the SIGIS project (see text) under which we conducted this study. We are also profoundly indebted to all of those in Ardmore who gave generously of their time to help us with it. And we would like to thank SIGIS colleagues, Roddy Ferguson and two anonymous referees for their helpful comments on earlier versions of this article.

Notes

1 An audit of ICT initiatives associated with Social Inclusion Partnerships and the Pathfinder Initiative was commissioned by the Central Research Unit of the Scottish Executive in December 1999, and captures the policy framework at the time (see http://www.scotland.gov.uk/cru/kd01/red/ audit01.htm, accessed March 2005). See Holligan (2003) for a more recent overview of a wider range of initiatives introduced under this framework.

2 The work of the Community Development Forum has been particularly significant in the UK – both in diffusing the model of community networks in the late 1990s (notably INSINC, 1997) and in sustaining a stream of research on them (e.g. Day and Harris, 1997; Harris and Dudley, 2000).

3 This policy framework can be critiqued on the grounds that digital inclusion is no panacea for social exclusion (Henwood *et al*, 2001). It should also be noted that the very terms digital inclusion and information society mean different things to different people and have been contested (see, e.g., Thomas and Wyatt, 2000). But these matters are not our concern here.

4 These same elements were flagged in another public-sector digital-inclusion initiative we studied for SIGIS, the National Opportunities Fund ICT Training for Teachers and School Librarians (Faulkner and Stewart, 2003). This understanding is consistent with economic and sociological studies of technology that highlight the necessarily context-dependent and 'embodied' character of technological capability (Faulkner, 1994).

5 Dudley and Harris (2001) is one of the few studies which addresses the relative success of different community-based ICT initiatives with specific user groups, though it does not offer a gender analysis of these differences.

6 The SIGIS project was funded by the Information Science and Technology section of the European Union under the Fifth Framework Programme, project number IST-2000-26329, and involved research teams from five European countries. Further information can be found on http://www.sigis.ist.org. The full reports of our case study of the Ardmore Network are in Kleif and Faulkner (2003) and Faulkner and Kleif (2004).

7 Accordingly, the evidence we present applies to that period only. The activities and uptake of the Ardmore Network have grown and evolved since then. However, we use the present tense throughout this article to preserve a sense of 'actuality'.

8 As an indication, the log for one afternoon at one village-based centre lists two mothers with young children who dropped in for help booking their holidays, three children from the local bike club, four hotel staff and a tourist wanting to use email, and a local reflexologist wanting help to design a website.

9 Although we are not in a position to assess the impact of the Ardmore Network on community development, it is clear that in some localities the Community Resource Centres have acted as a focus and catalyst for community activities. In line with our later discussion about informal networks of ICT expertise, our study indicates that targeting the Community Resource Centres on groups as well as individuals can have a multiplier effect in terms of ICT capability-building.

10 See our case study of vocational ICT training at the Edinburgh Women's Training Centre (Faulkner and Kleif, 2003; Kleif and Faulkner, 2004) and Harris and Dudley's (2000) study of five community-based Learning Centres.

11 In addition to the features mentioned here, the initiative is heterogeneous in other ways: The computers are backed up by technical support from a local contractor which means that people can rely on the technology to 'do the job'; were this not the case, users would soon loose faith in the Centres and stop coming. In this and other areas (e.g., the community web design), the network staff act as intermediaries between the users and technical experts.

12 In no way is this intended to imply that community networks have nothing to offer the under-30s or should not be trying to reach them. Rosie, who we meet below, is a classic illustration of the benefits a very young woman can gain from Community Resources Centres, and of the social exclusion reasons why particular groups of young people (in this case, the unemployed and mentally ill) should be targeted.

13 Counter to the normal presumption of computer enthusiasts being men, for two of these cases (both women) it was their mother who got them started.

14 One of the Resource Centres was staffed by a man for the first couple of years. We have evidence that he was effective in drawing in women (it was he who first showed Dorothy that she could visit art galleries on the Internet), but we are not able to assess whether he attracted more men than his female counterparts.

15 Admittedly, such diversity would be hard to achieve in a setting where there is only one Support Worker covering a large catchment area. None the less, having a diverse range of staff across a network would increase the likelihood of it generating the necessary range of strategies to effectively target all groups in the community (see below).

16 Although this general point is a constant refrain in contemporary gender studies, its full policy relevance is rarely grasped. Both public and private sector initiatives we studied as part of the SIGIS project were dogged by appeals to gender stereotypes which undermine their inclusion objectives, either by reinforcing unhelpful assumptions about gender and technology or by missing the large numbers of women and men who don't fit the stereotypes (Rommes *et al*, 2003; Sørensen, 2004).

References

Ascherson, N (2003) 'Designing virtual citizens: some Scottish experiments with electronic democracy', *Scottish Affairs*, No. 43 (Spring), pp 1–42.

Bijker, W E, Hughes, T P, and Pinch, T J (1987) *The Social Construction of Technological Systems*, Cambridge, MA: MIT Press.

Central Research Unit, Scottish Executive (1999). 'Audit of ICT initiatives', in *Social Inclusion Partnerships and Working for Communities Pathfinders in Scotland*, available online at http://www.scotland.gov.uk/cru/kd01/red/audit01.htm [accessed February 2005]

Clarke, A (2002) *Online Learning and Social Exclusion*, Leicester: NIACE.

Day, P and Harris, K (1997) *Down-to-Earth Vision: Community-based IT initiatives and Social Inclusion*, London: Community Development Forum and IBM UK.

Digital Scotland Task Force Report (2000) available online at http://www.scotland.gov.uk/Topics/Government/Open-scotland/17820/15760 [accessed February 2005]

Dudley, M and Harris, K (2001) *Pioneering Community Online Centres: A Management Perspective, A Report to the DfES*, London: Community Development Forum.

Faulkner, W (1994) 'Conceptualizing knowledge used in innovation', *Science, Technology, & Human Values*, vol 19(4), pp 425–58.

Faulkner, W and Kleif, T (2003) 'Edinburgh Women's Training Course: An old idea still working', in M Lie and KH Sørensen (eds) *Strategies of Inclusion: Gender in the Information Society, Vol I: Experiences from public sector initiatives*, Trondeim Norway: Centre for Technology and Society, NTNU, pp 201–36.

Faulkner, W and Kleif, T (2004) 'Included women, excluded men: Users and non–users of rural community resource centres', in N Oudshoorn, E Rommes, and I van Slooten (eds), *Strategies of Inclusion: Gender in the Information Society, Vol III: Surveys of Women's User Experience*, Trondeim, Norway: Centre for Technology and Society, NTNU, pp 137–66.

Faulkner, W and Stewart, J (2003), 'National ICT training for teachers: A case of "gender blindness"?', in M Lie and KH Sørensen (eds) *Strategies of Inclusion: Gender in the Information Society, Vol I: Experiences from public sector initiatives*, Trondeim, Norway, Centre for Technology and Society, NTNU, pp 273–97.

Gurstein, M (2000) *Community Informatics: Enabling communities with information and communication technologies*, USA: Idea Publishing Group.

Håpnes, T and Rasmussen, B (1991) 'Excluding women from the technologies of the future?', *Futures*, December, pp 1107–19.

Harding, S (1986) *The Science Question in Feminism*, Ithaca, New York: Cornell University Press.

Harris, K and Dudley, M (2000) *Many First Steps: Interim report to DTI and DfEE Learning Centre Board on the first phase of 'the five' pioneer Learning Centres*, available online at http://www.riservices.co.uk/Manyfirststeps.htm [accessed February 2005]

Henwood, F, Kennedy, H and Miller, N (eds) (2001) *Cyborg Lives? Women's Technobiographies*, York: Raw Nerve Books.

Holligan, C (2003) 'View from the cyberfloor: Modes of social inclusion in Scotland', *Scottish Affairs*, vol 43 (Spring), pp 87–107.

INSINC (1997) *The Net Result: Report of the National Working Party on Social Inclusion in the Information Society*, London, IBM UK and Community Development Forum.

Kleif, T and Faulkner, W (2003) 'Rural community resource centres: A case of "de facto" women's inclusion in the information society', in M Lie and KH Sørensen (eds), *Strategies of Inclusion: Gender in the Information Society, Vol I: Experiences from public sector initiatives*, Trondeim,

Norway: Centre for Technology and Society, NTNU, pp 237–72.

Kleif, T and Faulkner, W (2004) '"I am not the same person!" User study of a women-only training course', in N Oudshoorn, E Rommes, and I van Slooten (eds), *Strategies of Inclusion: Gender in the Information Society. Vol III: Surveys of Women's User Experience*, Trondheim, Norway: Centre for Technology and Society, NTNU, pp 107–36.

Laurie, N, Dwyer, C, Holloway, S L and Smith, F M (1999) *Geographies of New Femininities*, New York: Longman.

Lohan, M and Faulkner, W (2004) 'Masculinities and technologies: Some introductory remarks', *Men and Masculinities, Special Issue on Masculinities and Technologies*, vol 6 (4), pp 319–29.

Malina, A (2001) ' Electronic community networks: An examination of community development practice, e-democracy and community learning', *Journal of Community Work and Development*, vol 1(2), pp 68–80.

McGivney, V (1998) *Excluded Men: Men who are missing from education and training*, Leicester: NIACE.

McGivney, V (2000) *Recovering Outreach: Concepts, practices and issues*, Leicester: NIACE.

MacKeogh, C (2003) 'The Cork Institute of Technology and Fastrack to IT Initiatives for the lone parents and the long-term unemployed', in M Lie and KH Sørensen (eds), *Strategies of Inclusion: Gender in the Information Society, Vol. 1: Experiences from public sector initiatives*, Trondheim, Norway: Centre for Technology and Society, NTNU, pp 419–38.

Midgely, S (1997) 'Putting the IT in community', *FE Now*, June.

Miller, D (1999) 'Community Information Networks' in S Pantry (ed), *Building Community Information Networks: Strategies and experiences*, London: Library Association.

OFSTED (2002) *ICT in Schools: Effect of government initiatives*, Report produced by the Office for Standards in Education, April 2002, London: The Stationery Office.

Pantry, S (ed) (1999) *Building Community Information Networks: Strategies and experiences*, London: Library Association.

Policy Action Team 15 (2002) *Policy Action Team 15 Report: Closing the digital divide*, London, UK Social Inclusion Unit.

Rommes, E, van Slooten, I,van Oost, E, Oudshoorn, N (2003) (eds) *Designing Inclusion: The development of ICT products to include women in the Information Society*, available online at http://www.rcss.ed.ac.uk/sigis/public/deliverables/D06/1 [accessed February 2005].

Schuler, D (1996) *New Community Networks: Wired for change*, New York: Addison-Wesley.

Sørensen, K H (2004) *Gender and Inclusion Policies for the Information Society*, available online at http://www.rcss.ed.ac.uk/sigis/public/deliverables/D07/1 [accessed February 2005].

Sørensen, K H and Stewart, J (eds (2002) *Digital Divides and Inclusion Measures: A review of literature and statistical trends on gender and ICT*, Trondheim, Norway: Centre for Technology and Society, NTNU, December.

Stewart, J (2001) 'Encounters with the Information Society', Unpublished PhD thesis, Edinburgh: The University of Edinburgh.

Tett, L (1994) 'Where have all the men gone? Adult participation in continuing education', *Scottish*

Journal of Adult and Continuing Education, vol 1(2), pp 41–49

Wyatt, S and Thomas, G (2000) 'Access is not the only problem: using and controlling the Internet', in W Wyatt, F Henwood, N Miller and P Senker (eds), *Technology and In/equality: Questioning the Information Society*, London: Routledge.

Wyatt, S, Thomas, G and Terranova, T (2002) 'They came, they surfed, they went back to the beach: Conceptualising use and non–use of the Internet', in S Woolgar (ed), *Virtual Society? Technology, Cyberpole, Reality*, Oxford: Oxford University Press, pp 23–40.